

DATA SCIENCE & ANALYTICS INTERNSHIP | ZIDIO DEVELOPMENT

Retail Analytics Dashboard

Interactive Business Intelligence dashboard for sales, customer, product, geographic and profitability analysis

HTML5 + CSS3 + JavaScript

Chart.js v4.5.1

Python pandas numpy

Data Science & Analytics Internship Project

Zidio Development

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Data Science & Analytics Intern | MCA – Data Science | 2026

TOTAL REVENUE

\$14.09M

TOTAL PROFIT

\$3.34M

PROFIT MARGIN

23.7%

TOTAL ORDERS

23,960

UNIQUE CUSTOMERS

6,072

AVERAGE ORDER VALUE

\$668



Presentation Roadmap

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- Introduction
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Abstract



Purpose

Static, browser-based dashboard analyzing sales, customer behavior, product performance, geography, profitability and operational KPIs from a large synthetic retail dataset.



Architecture

Self-contained HTML file with embedded JSON data and Chart.js - no backend server or external runtime dependencies.



Capabilities

Executive KPIs, date-comparison controls, category and geographic drill-down filters, RFM segmentation, cohort analysis, forecasting and a transactional detail table.



Verification

Verified on a local web server: the page rendered with the expected overview metrics, filters and charts. Built from real project files, data values and source code - no fabricated KPI values.

23,960

Records after cleaning

\$14.09M

Total revenue

12

Analysis sections



Introduction

Why retail analytics?

- Turns revenue trend, product performance, customer loyalty and operational efficiency into one decision-making view.
- Interactive dashboard summarizes global retail performance from a large synthetic order dataset.
- Static file delivery: HTML, JavaScript, CSS and a pre-generated JSON data file - opens in a browser without a server-side application.
- Repository holds the dashboard interface, the data pipeline and the automation files that generate the final single-page dashboard.

12 ANALYSIS SECTIONS

1

Overview

2

Trends

3

Products

4

Customers

5

Segments

6

Geography

7

Profitability

8

Operations

9

Advanced
Analytics

10

Data Quality

11

Insights

12

Detail Table



Problem Statement



The challenge

- Retail businesses hold large volumes of transactional data that are hard to interpret in raw spreadsheets alone.
- Managers need one consolidated view of revenue, profit margin, customer activity, channel performance and order status.
- Without a dashboard, performance is fragmented across multiple reports - harder to compare over time, by segment, geography or product category.



The solution

- Converts a large synthetic retail dataset into a structured, compact analytical dashboard.
- Business-ready and transparent: all major metrics are visible in the interface and filters are interactive.
- The report is derived from the original data rather than manually written static numbers.

From fragmented raw records to one interactive, decision-ready view of retail performance



Project Objectives



01

Interactive retail dashboard

Visualize transactional data in a clear, business-oriented interface.



02

Executive KPIs

Revenue, profit, margin, orders, valid orders, AOV, units sold, return and cancellation rate.



03

Flexible data slicing

Date range, category, sub-category, region, country, segment, channel and order status.



04

Customer behavior

Analyze customers through segmentation and cohort retention scoring.



05

Product & geographic analysis

Drill-down navigation and trend comparison.



06

Self-contained static page

Embedded analytics data and Chart.js library in one webpage.



07 Generate a professional academic project report based on the actual implementation and verified screenshots.



Technology Stack



Front-end



HTML5 & CSS3

Dashboard structure, styling and responsive layout.



Vanilla JavaScript

Interactive filtering, aggregation, rendering and chart configuration.



Chart.js v4.5.1

Vendored locally in the scripts folder; renders all dashboard charts.



Data & ETL



Python: pandas & numpy

Data generation, cleaning, validation, aggregation and forecasting logic.



JSON

Precomputed data exchange format between the ETL pipeline and the browser dashboard.



Runtime & Reporting



Local Python HTTP server

Verifies the dashboard runtime in the browser at <http://localhost:8000>.



ReportLab

Generates the final PDF report from verified project data and screenshots.



Dataset Description

24,288

Raw rows (raw_sales.csv)

23,960

Rows after cleaning

12

Countries

5

Regions

1 Jan 2022
30 Aug 2025

Date range covered

Dataset columns (19)

Time

order date

ship date

Product

product name

category

sub-category

quantity

unit price

discount

Financial

sales

cost

profit

shipping cost

Customer & place

customer ID

region

country

city

Channel & status

payment method

channel

order status



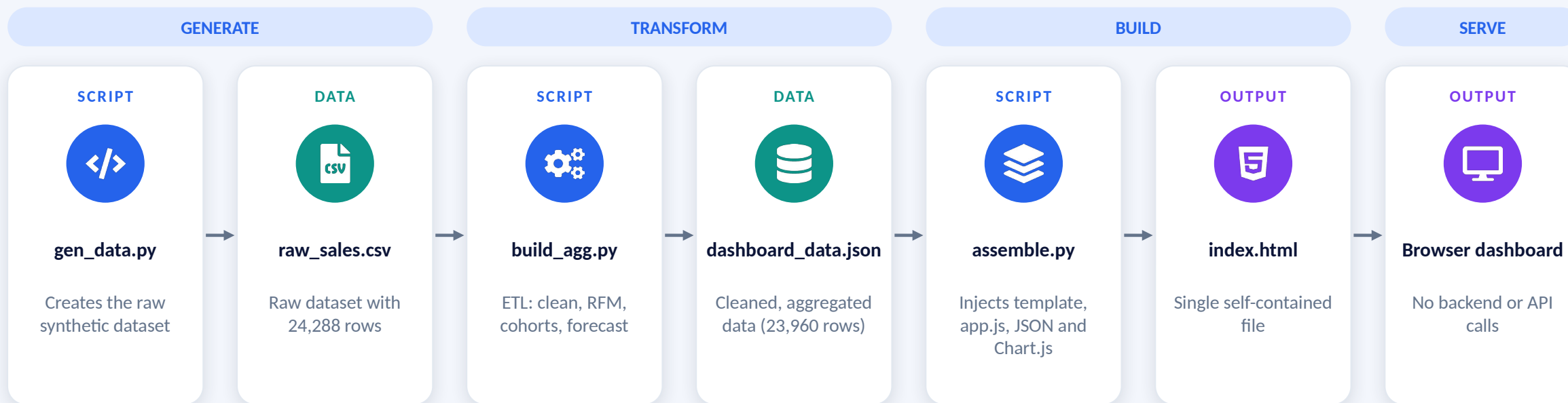
Intentional data issues

- Missing values
- Duplicate rows
- Sign errors
- Inconsistent text casing
- Statistical outliers

Synthetic data built for demonstration; data-quality checks reflect real issues, not staged values.



System Architecture



Static data-pipeline pattern

The generation script builds the raw dataset; the aggregation script cleans and enriches it; the browser consumes the final page.



One self-contained file

`index.html` embeds the chart library and JSON - open it directly or serve it locally, with no build process.



Repository layout

Dashboard at the root; assets/images, data, scripts and screenshots in dedicated folders.



Project Folder Structure

```
retail-analytics-dashboard/  
├── index.html  
├── README.md  
├── LICENSE  
├── .gitignore  
├── assets/  
│   └── images/  
│       └── favicon.svg  
├── data/  
│   ├── dashboard_data.json  
│   └── raw_sales.csv  
├── scripts/  
│   ├── app.js  
│   ├── assemble.py  
│   ├── build_agg.py  
│   ├── gen_data.py  
│   ├── template.html  
│   └── vendor/  
│       └── chart.umd.min.js  
└── screenshots/  
    └── README.md
```



index.html

Dashboard page at the project root.



assets/

Asset files under images (favicon.svg).



data/

raw_sales.csv and dashboard_data.json.



scripts/

Building and processing logic: app.js, assemble.py, build_agg.py, gen_data.py, template.html and vendor/chart.umd.min.js.



screenshots/

Verified dashboard screenshots.



Implementation & Code Workflow



scripts/gen_data.py

Data generation

- Creates the synthetic retail dataset
- Writes data/raw_sales.csv
- Includes intentional data issues: missing values, duplicates, sign errors, inconsistent casing and outliers



scripts/build_agg.py

ETL processing

- Reads the raw CSV
- Checks missing, invalid and duplicate records
- Standardizes values; computes quality metrics
- Calculates RFM scores
- Builds cohort retention tables
- Generates forecast metrics and writes data/dashboard_data.json



scripts/assemble.py

Assembly

- Reads the HTML template, app.js, JSON dataset and vendored Chart.js
- Injects them into index.html
- Result is self-contained - no runtime fetching of data or libraries



scripts/app.js

Dashboard logic

- State management and filtering logic
- Tab rendering, KPI calculations, chart building
- Table generation and detail-table interactions
- Global filters, period and comparison modes
- Drill-downs: product, customer, geographic, profitability



Executive KPIs



TOTAL REVENUE

\$14.09M

TOTAL PROFIT

\$3.34M

PROFIT MARGIN

23.7%

TOTAL ORDERS

23,960

VALID ORDERS

21,075

88.0% of total



AVERAGE ORDER VALUE

\$668

UNIQUE CUSTOMERS

6,072

UNITS SOLD

33,893

RETURN RATE

6.0%

CANCELLATION RATE

6.1%

● Financial

● Operational

● Customer

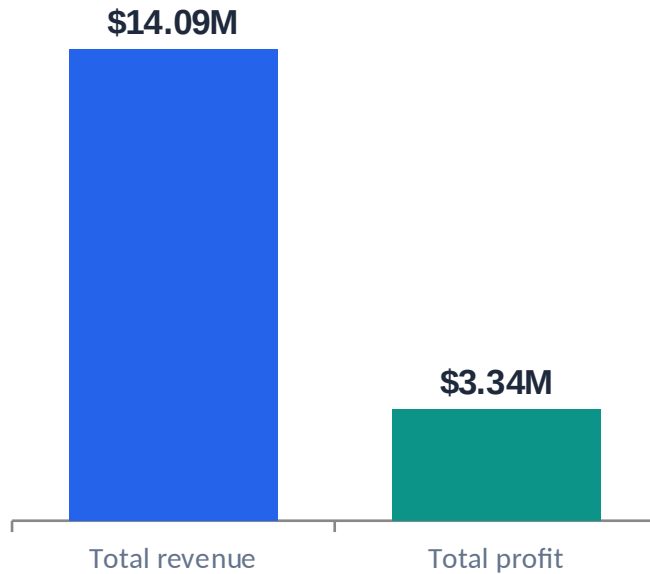
Values read from the running dashboard (Custom Range: 01 Jan 2022 to 30 Aug 2025).



Results at a Glance

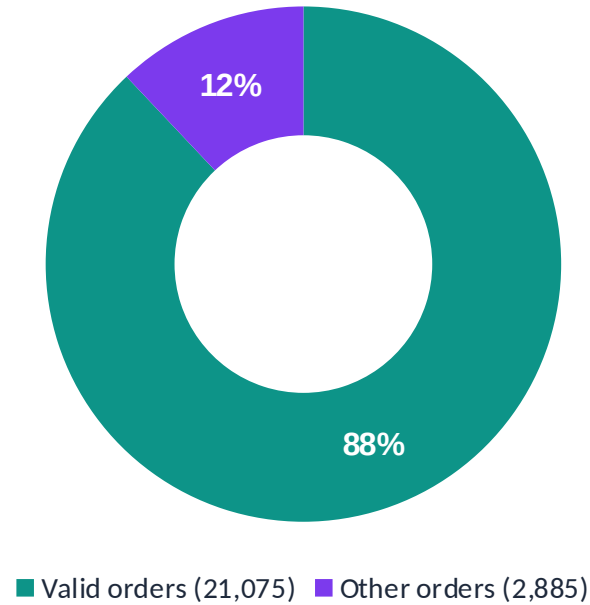
Revenue vs. profit

USD millions



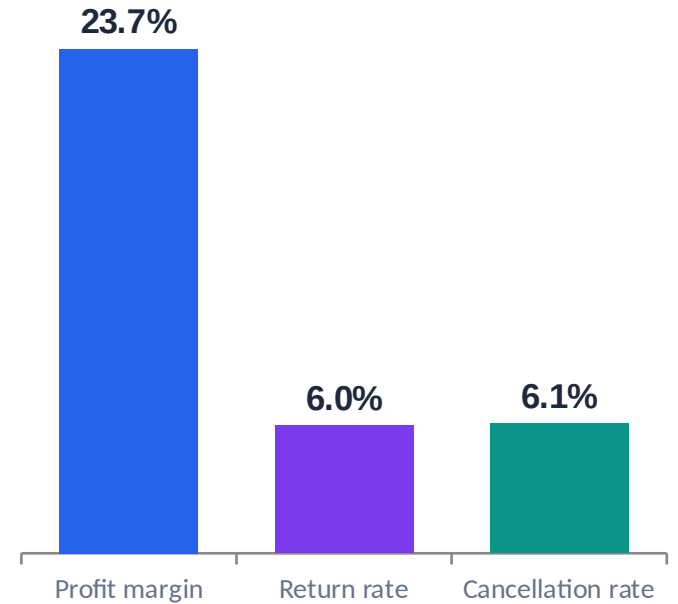
Order validity

Valid vs. other orders



Key rates

Percent



Margin: \$3.34M profit / \$14.09M revenue = 23.7%

Valid orders: 21,075 = 88.0% of 23,960 total

6,072 customers | 33,893 units | \$668 AOV



Dashboard Features



Global date filters

Start and end date controls with a Reset All action.



Period controls

Custom Range, MTD, QTD and YTD comparisons.



Comparison modes

Previous Period and Previous Year overlays.



Overview KPI cards

Financial, operational and customer KPIs.



Product analytics

Category-to-sub-category drill-down logic.



Customer analytics

RFM segmentation and cohort retention heatmap.



Geographic analysis

Region, country and city views.



Operations & forecasting

Order status and processing views; forecasting and advanced analytics.



Quality & detail table

Data quality monitoring; detail table with pagination and filtering.



Data Processing / ETL Workflow

1

Generate

`gen_data.py`

- Builds the synthetic raw_sales.csv
- Deliberately includes imperfect records

2

Clean

`build_agg.py`

- Missing values
- Duplicate rows
- Invalid quantities
- Zero unit prices
- Raw outliers
- Inconsistent category casing
- Inconsistent or missing customer-level metadata

3

Assess quality

`build_agg.py`

- Quality metrics on raw data
- Column status: Good, Minor Issues, Needs Attention
- Derived fields: profit, shipping cost, processing days
- Completeness, validity and uniqueness scores feed the data-quality cards

4

Enrich & forecast

`build_agg.py`

- Customer RFM: recency, frequency, monetary
- Cohort retention tables from first-purchase dates
- Daily revenue forecast using a linear trend fit

5

Serialize & embed

`assemble.py`

- Output file: dashboard_data.json, saved in data/
- Embedded into the final index.html



Charts & Visualizations



Overview

Category revenue, top regions and order-status mix charts.



Products

Category or sub-category performance.



Trends

Metric overlays across day, week, month, quarter and year granularity.



Advanced analytics

RFM bar charts, cohort retention heatmaps and forecast charts with confidence bands.



Profitability & operations

Profitability views and operational status charts.



Detail table

Transaction inspection with page-size control, search, sorting and CSV export-oriented behavior.

Trend metric overlays

Revenue

Profit

Orders

Customers

Margin

Time granularity

Day

Week

Month

Quarter

Year

All charts are drawn with Chart.js v4.5.1 and support both high-level decision-making and drill-down analysis.



Testing & Results

1

Start a local server from the project directory.

2

Load the page in a browser at <http://localhost:8000>.

3

Browser reported the title: Global Retail Analytics Dashboard | Data Analytics Portfolio.

4

Visible: title, filter bar, KPI cards and initial overview content.



Page loaded successfully



Embedded JSON and Chart.js assets served correctly



Expected aggregate values and filter controls shown



Static design: no external API or server-side app required

This is the project's verified runtime state - not an assumed or hypothetical result.



Screenshot: Running Dashboard

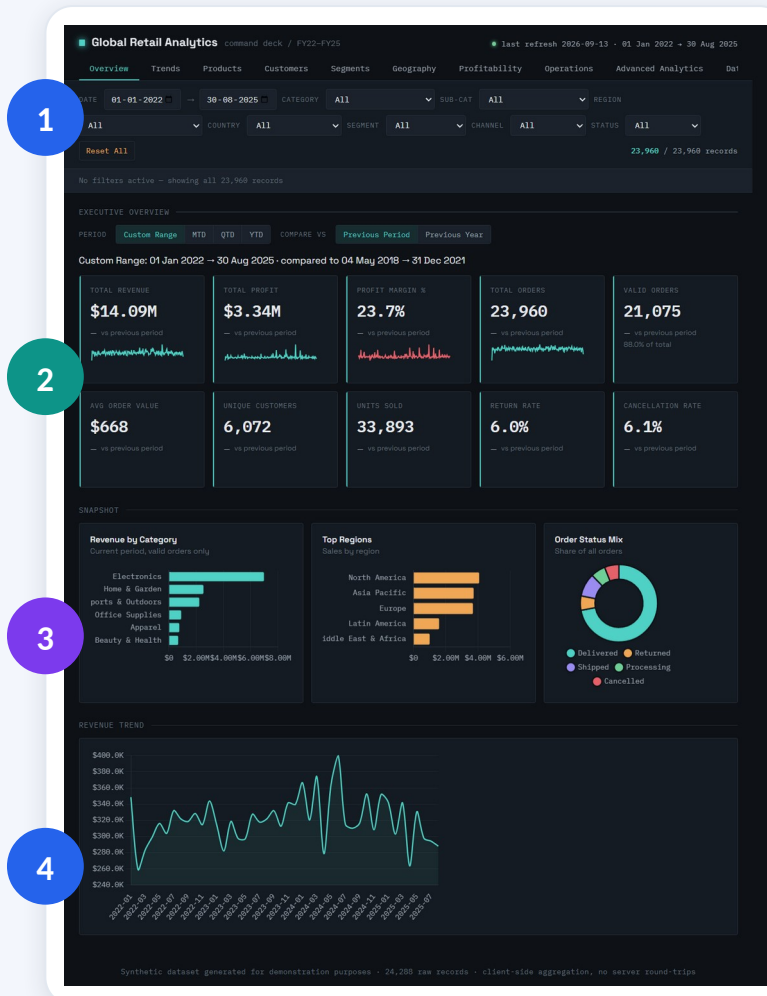


Figure 1 - Screenshot of the Actual Running Dashboard

Overview tab | 01 Jan 2022 to 30 Aug 2025 | 23,960 / 23,960 records

1

Header, tabs & global filters

Date, category, sub-category, region, country, segment, channel and status filters with Reset All; no filters active, all 23,960 records shown.

2

Executive overview KPI cards

Period (Custom Range, MTD, QTD, YTD) and compare (Previous Period, Previous Year) controls above ten KPI cards.

3

Snapshot charts

Revenue by Category, Top Regions and Order Status Mix (Delivered, Returned, Shipped, Processing, Cancelled).

4

Revenue trend

Line chart across the full date range. Footer: synthetic dataset, 24,288 raw records, client-side aggregation, no server round-trips.



Screenshot Close-ups

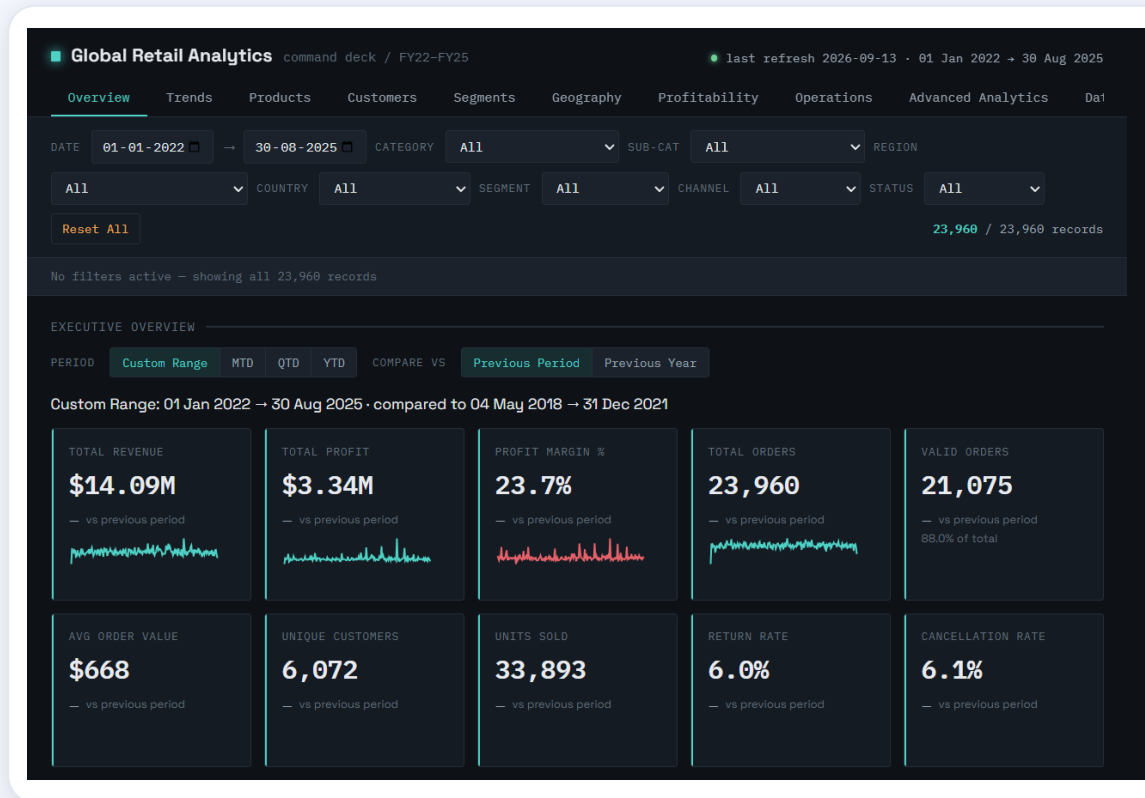


Figure 2 - Filter bar and executive KPI cards

Global filters, period and comparison controls, and the ten KPI cards.

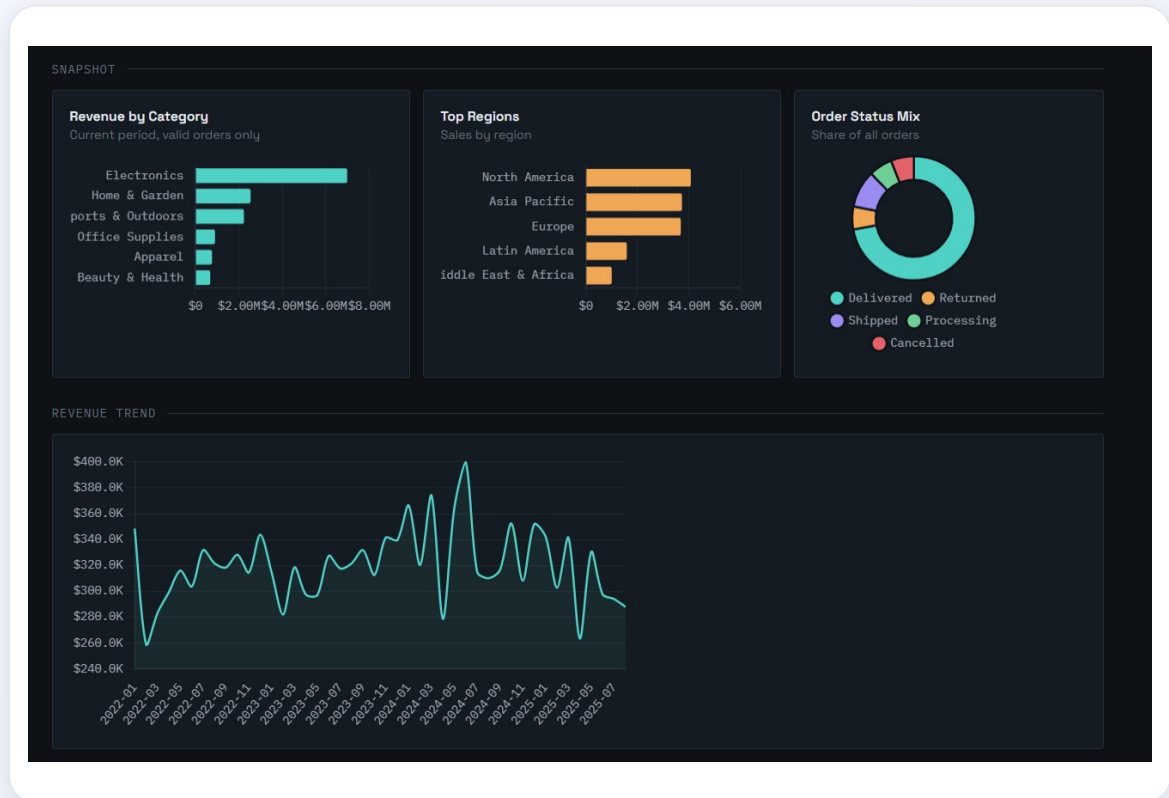


Figure 3 - Snapshot charts and revenue trend

Revenue by Category, Top Regions, Order Status Mix and the revenue trend line.



Challenges & Solutions

CHALLENGE



Synthetic data had missing values and metadata, duplicate records, invalid quantities and unusual outliers.



The dashboard needed to work without a backend or external runtime.



Large datasets are cumbersome to display in a browser.



Academic and portfolio reporting requires clear business interpretation.



SOLUTION



ETL standardizes and cleans values while tracking quality metrics and statistical anomalies.



A single self-contained HTML file with inlined data and Chart.js.



Pre-aggregated data with report-friendly client-side logic, filtered rendering and paginated table operations.



Metrics organized into executive KPI cards, analytical tabs and clear chart labels.



Future Enhancements



01

Live data refresh

Server-side or edge-cached data refresh for live, non-synthetic data updates.



02

Geographic map view

A true map view for region-level and country-level analysis.



03

State / province geography

Extend to state or province level if such data becomes available.



04

Advanced forecasting

Replace the current linear forecast with a seasonal or machine-learning model.



05

Shareable filter presets

Support URL-based filter presets.



06

GitHub Pages deployment

Extend the report engine and dashboard packaging for GitHub Pages.



Conclusion

Data generation



ETL



Metric computation



Analytical visualization



Static presentation



What was delivered

An interactive, business-focused dashboard built from a realistic synthetic retail dataset - with actual KPI values, interactive filters, advanced analytics and a data-quality narrative.



Why it qualifies

Suitable for an MCA or Data Analytics submission: real project source code, actual dataset logic, verified runtime testing and a coherent business analysis narrative.



The outcome

A clean, browser-based dashboard with no backend dependencies - functional, evidence-backed and with a clear path for future enhancement.



GitHub & Live Demo Links



Source Code / GitHub

<https://github.com/SaurabhPatil-393/retail-analytics-dashboard>

Complete project source code, data, scripts and documentation.



Live Dashboard

<https://saurabhpatil-393.github.io/retail-analytics-dashboard/>

Interactive Retail Analytics Dashboard.



Portfolio

<https://saurabhpatil-393.github.io/portfolio>

Professional portfolio and additional projects.

Thank You

Questions & discussion

Retail Analytics Dashboard

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